

Notes: Harris and Gurel (JF, 1986)

Price and Volume Effects Associated with Changes in the S&P 500 List: New Evidence for the Existence of Price Pressures

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1 Big picture

- **Question.** Do large, non-information-motivated demand shifts move stock prices, and if they do, are the effects temporary *price pressures* or permanent changes in equilibrium prices?
- **Setting.** The paper studies additions to and deletions from the S&P 500 list between 1973 and 1983. These events are attractive because they trigger buying and selling by index funds, yet Standard & Poor's selection rules are not supposed to reflect private information about future returns.
- **Core challenge.** A large sale or purchase can move price for at least three very different reasons: information about fundamentals, imperfect substitution across securities, or short-run liquidity pressure. Most earlier event settings such as block trades and secondary offerings are contaminated by information effects.
- **Main conceptual contribution.** The paper treats S&P 500 composition changes as close to *information-free demand shocks*. That lets the authors separate temporary price pressure from permanent information or redistribution effects more cleanly than in prior studies.
- **Competing hypotheses.**
 - **EMH / perfect substitutes:** if the trade contains no new information, large trades should occur near the existing market price.
 - **Imperfect substitutes hypothesis (ISH):** demand curves slope downward even in the long run, so a demand shift should change the equilibrium price permanently.
 - **Price-pressure hypothesis (PPH):** short-run liquidity suppliers require compensation to absorb an unexpected demand shock, so price moves immediately but later reverses toward the full-information level.
- **Main empirical design.** The authors examine event-time trading volume and event-study excess returns around S&P 500 additions, with special attention to how the results differ between 1973–1977 and 1978–1983, the period when index funds became large.
- **Main baseline result.** For additions, trading volume jumps sharply after the announcement and prices rise immediately. In the later subperiod 1978–1983, the average excess return is about 3.13% on day 1 and 2.77% over days 1 to 5.
- **Main reversal result.** The day-1 price increase is substantially reversed over the next two to three weeks. In 1978–1983, cumulative excess return from day 2 to day 21 is about -2.49% , and the paper's formal tests favor *full reversal* over *no reversal*.
- **Main credibility result.** The paper argues the announcement itself conveys little if any new information: S&P says investment appeal does not enter its decisions, the price effect appears only when index funds become large, pre-announcement returns and volume show little anticipation, and the initial price increase reverses.
- **Supporting breadth result.** A small clean sample of deletions also shows a negative day-1 price effect and a large volume response, though the sample is too small to draw strong conclusions about reversal dynamics.
- **Economic takeaway.** The evidence supports temporary price pressure rather than the view

that these demand shocks permanently raise fundamental values. The paper is therefore one of the early clean demonstrations that large trades can move prices even without new information.

2 Data and measurement (key objects)

2.1 Why S&P 500 list changes are a useful laboratory

- S&P 500 changes are typically triggered by events such as mergers, tender offers, bankruptcy, firms becoming too small, or S&P's desire to keep the list representative of the U.S. market.
- S&P explicitly states that judgments about a stock's investment appeal do *not* enter the selection process.
- These changes therefore should not reveal private information about future return distributions, at least relative to most event settings used in earlier price-impact studies.
- At the same time, S&P 500 index funds are forced to rebalance: when a firm is added, index funds buy it; when a firm is deleted, they sell it.

Interpretation. The paper's core identification idea is that S&P changes create a shift in demand without obviously revealing information about fundamentals.

2.2 Why index-fund demand can be large

- By the end of 1983, public index funds held about \$35.7 billion, equal to roughly 2.96% of the market value of the S&P 500.
- The authors argue that private in-house index funds probably held about the same amount again.
- If both public and private index funds bought an added security, demand for that security could rise by about 5.92%.

Interpretation. Even if the S&P announcement itself is information-free, it can still generate a large order imbalance. That is exactly the environment in which price pressure should show up.

2.3 Sample construction and announcement timing

- The full universe of changes from 1973 to 1983 contains 228 additions and 228 deletions.
- The main additions sample excludes 34 cases: 21 where the addition was linked to a merger with a deleted firm, and 13 where the added stock traded over the counter.
- That leaves 194 additions for the main event-study sample.
- Most deletion events are not clean because the deleted security often disappears in a merger, tender offer, or bankruptcy, so the paper focuses mainly on additions.

Announcement timing is handled carefully:

- Before September 1976, changes were first published in *Standard and Poor's The Outlook*, which arrived on Mondays; the Monday arrival date is treated as the first public announcement date.

- Starting on September 22, 1976, S&P began a Notification Service announcing changes on the day of the change, always a Wednesday after the close; that date is taken as the event date for the later sample.

Interpretation. The split in announcement technology matters because precise event dating is essential when the paper studies day-1 price and volume responses.

2.4 Volume measure

For security i in event-time period t , the paper defines a standardized volume ratio:

$$MVR_t = \frac{1}{N} \sum_i V R_{it}, \quad (1)$$

where

$$V R_{it} = \frac{V_{it}}{V_{mt}} \frac{V_{m\cdot}}{V_{i\cdot}}. \quad (2)$$

Here:

- V_{it} is trading volume of security i in event-time period t ,
- V_{mt} is total NYSE trading volume in event-time period t ,
- $V_{i\cdot}$ and $V_{m\cdot}$ are average security and market volume over the 8 weeks preceding the announcement week.

Interpretation. The expected value of $V R_{it}$ is 1 if the announcement has no effect on trading volume. Market adjustment is used to remove aggregate variation in trading activity.

2.5 Excess-return measure

The paper uses a standard market-model event study:

$$R_{it} = \alpha_i + \beta_i R_{mt} + u_{it}, \quad (3)$$

and defines the excess return as

$$ER_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt}). \quad (4)$$

Key implementation details are:

- The market model is estimated using CRSP daily returns from 250 days before the announcement to 40 days after it, excluding days 0 to 10.
- The paper mainly reports results using the CRSP value-weighted index, though the equal-weighted index gives nearly identical conclusions.
- To control for heteroskedasticity, the event-day excess returns are standardized by the root-mean-squared error from the market-model regression when forming t -statistics.

Interpretation. The excess-return setup asks whether additions earn abnormal returns relative to normal co-movement with the overall market.

2.6 Cumulative reversal measure

To study how much of the day-1 price rise is later reversed, the paper defines cumulative excess returns from day 2 to day T :

$$CER_{iT} = \sum_{\tau=2}^T ER_{i\tau} \quad (\text{implemented as cumulative geometric excess returns in the paper}). \quad (5)$$

The late-sample reversal model is written as

$$ER_{i1} = \mu_i + \varepsilon_i, \quad (6)$$

$$CER_{iT} = \phi_i + \eta_i, \quad (7)$$

with

$$\varepsilon_i \sim N(0, \sigma^2), \quad \eta_i \sim N(0, (T-1)\sigma^2), \quad (8)$$

independently.

The paper compares:

$$H_0 : \phi_i = 0 \quad \text{for all } i \quad (\text{no reversal}), \quad (9)$$

$$H_A : \phi_i = -\mu_i \quad \text{for all } i \quad (\text{full reversal}). \quad (10)$$

Interpretation. This is the paper's clean test of ISH versus PPH. No reversal favors permanent demand effects; full reversal favors temporary price pressure.

2.7 Subperiod splits and why they matter

- The paper reports results for the full 1973–1983 sample, but also splits the sample into 1973–1977 and 1978–1983.
- This split is economically important because index funds were tiny in the early years and much larger by the late 1970s and early 1980s.
- Table II illustrates this using the growth of two major public index funds: the Wells Fargo S&P Fund rises from \$170 million in 1974 to \$6.13 billion in 1983, and the Vanguard fund rises from \$14 million in 1976 to \$234 million in 1983.

Interpretation. If the event effect truly comes from index-fund demand, it should be much stronger in the later subperiod. The paper treats this as an important credibility check.

2.8 No-information evidence collected by the paper

The paper assembles three kinds of evidence that S&P additions are not information events:

- S&P's stated criteria exclude investment appeal.
- Before the Notification Service began, only about 5 to 10 requests per year for early listing information were made to S&P's director of index services, mostly by index fund managers rather than information traders.

- After the Notification Service started, the early subscriber base was again dominated by index fund managers.

Interpretation. If S&P additions really conveyed valuable information about future stock returns, one would expect much stronger demand for advance knowledge from information-motivated traders.

2.9 The deletion sample

- A clean deletions sample is much smaller. In 1978–1983 only 13 deletions are usable after excluding cases caused by merger, bankruptcy, or tender offer.
- Six of those 13 happen on November 30, 1983 when utilities were deleted to make room for the new Bell operating firms, so clustering is also a concern.

Interpretation. The deletion evidence is useful as a sign check, but the paper is right to rely mainly on additions for the stronger inference.

2.10 Descriptive facts that matter for the paper

- In the full sample of additions, average day-1 volume is 1.89 times normal and average day-1 excess return is 1.52%.
- In 1978–1983, the corresponding numbers are much larger: day-1 volume ratio of 2.81 and day-1 excess return of 3.13%.
- Pre-announcement volume in 1978–1983 is close to normal for days -2 to 0 , and pre-announcement excess returns are mostly near zero.

Interpretation. The late-sample pattern already looks like what price pressure predicts: a sharp immediate demand shock, a large day-1 price move, and little sign that the move was anticipated long before the announcement.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing hypotheses

The paper organizes the evidence around three theoretical views.

- **Efficient market hypothesis (EMH).** Securities are close substitutes and long-run as well as short-run demand is highly elastic. A large trade with no information should not move price much.
- **Imperfect substitutes hypothesis (ISH).** Securities are not close substitutes even in the long run. A demand shift changes the equilibrium distribution of holdings, so the price effect should be permanent.
- **Price-pressure hypothesis (PPH).** Long-run demand is elastic at the full-information price, but short-run liquidity suppliers require compensation for inventory risk and transaction costs.

Price therefore moves immediately and later reverses.

Interpretation. The key empirical distinction is not merely whether prices move, but whether the movement is temporary or permanent.

3.2 Volume-ratio specification

The paper’s first empirical question is whether S&P additions trigger abnormal trading. The cross-sectional mean standardized volume ratio,

$$MVR_t = \frac{1}{N} \sum_i VR_{it}, \quad (11)$$

serves as the test statistic.

Interpretation. A significant rise in MVR_t is the paper’s first sign that the event is truly creating a demand shift rather than being only a bookkeeping reclassification with no trading implications.

3.3 Event-study price specification

The second empirical question is whether prices respond abnormally after the announcement. The paper calculates mean excess returns on day 1 and over days 1 to 5.

Interpretation. A positive day-1 excess return following additions is consistent with either PPH or ISH under a no-information interpretation. It is the *reversal evidence* that distinguishes the two.

3.4 Reversal tests

The paper runs two complementary tests of reversal:

- a two-sample t -test of whether the mean cumulative excess return from day 2 to day T equals the negative of the mean day-1 excess return;
- a Bayesian odds-ratio comparison between no reversal and full reversal using the model above.

Interpretation. These are much stronger than simply noting that post-event returns are often negative. The authors ask directly whether the initial abnormal price increase is *fully corrected*, as PPH predicts.

3.5 Pre-announcement and post-announcement event windows

For the late subperiod the paper studies daily event time from day -10 to day 30 and weekly volume windows around the announcement.

- Because late-period announcements are made after the close on Wednesday, day 1 is Thursday.
- Week 0 contains days -2 to 2, and week 1 contains days 3 to 7.

Interpretation. This event-time structure lets the authors distinguish anticipation, immediate pressure, short-run reversal, and longer-run persistence in trading volume.

3.6 Opening-price decomposition

For a smaller transactions-data subsample from December 1981 to January 1983, the authors decompose the day-1 return into a close-to-open and open-to-close component.

Interpretation. This is not central to the paper, but it shows that a good part of the price adjustment happens between the announcement and the next market open, which is exactly what one would expect if the market quickly capitalizes the demand shock.

3.7 Deletions as a sign test

The paper does not build a full structural comparison on deletions because the sample is too small, but it still checks whether clean deletions are followed by higher volume and a negative price move.

Interpretation. This is a useful directional test: if additions push prices up through demand pressure, deletions should push them down.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no traditional instrument

The paper does not use an instrumental-variable design. Identification comes instead from a *clean event choice*: S&P additions look like demand shocks with little information content.

Interpretation. Credibility therefore depends on the plausibility of the no-information claim and on the temporary nature of the price response.

4.2 Why S&P changes are unusually credible demand shocks

The paper's strongest identifying arguments are:

- S&P says investment appeal is not part of the selection rule.
- Many changes are induced mechanically by merger, tender offer, bankruptcy, or representativeness concerns.
- Index funds create a direct and easily interpretable source of excess demand.

Interpretation. Relative to block sales or new issues, this setting is much closer to an exogenous demand shift that is separate from adverse or favorable inside information.

4.3 Why the no-information claim is empirically supported

The paper makes three empirical arguments.

- **Low demand for advance knowledge.** If S&P additions contained valuable information, information traders should have tried hard to obtain advance notice, yet very few requests were made and the early subscriber base to the Notification Service was dominated by index fund

managers.

- **Time-series pattern.** If the announcement conveyed information, the price effect should appear throughout the sample. Instead it is essentially absent in 1973–1977 and strong only in 1978–1983, exactly when index funds are large.
- **Reversal.** If the day-1 price rise reflected new information about fundamentals, the effect should persist. Instead it reverses strongly.

Interpretation. The argument is cumulative. No single piece is decisive, but together they make the information story hard to sustain.

4.4 Why the late-sample versus early-sample comparison is powerful

The paper’s period split is not arbitrary. The economic logic is:

- the information content of an S&P addition should not depend much on the size of index funds;
- the demand-shift effect should depend very strongly on the size of index funds;
- therefore a much larger effect in 1978–1983 than in 1973–1977 supports PPH and weakens the information story.

Interpretation. This is one of the cleanest design features in the paper because it turns time variation in the strength of index-fund demand into a built-in falsification test.

4.5 Why reversal discriminates sharply across theories

- **EMH with no information:** no abnormal price effect.
- **EMH with information:** abnormal price effect, but it should be permanent.
- **ISH:** abnormal price effect, and it should be permanent because the long-run demand curve slopes downward.
- **PPH:** abnormal price effect, followed by reversal as liquidity suppliers are compensated.

Interpretation. The reversal evidence is not a minor add-on. It is the central reason the paper concludes in favor of price pressure rather than imperfect substitution.

4.6 Pre-announcement evidence

The paper checks whether the market seems to anticipate the announcements.

- In 1978–1983, mean volume ratios for days -2 to 0 are insignificantly different from 1.
- Only one daily excess return in the pre-event window (day -10) is significantly different from zero.

Interpretation. The event is not obviously leaking into the market well before the official announcement date. That strengthens the interpretation of the day-1 reaction as a response to the

announcement itself.

4.7 Alternative interpretations the paper discusses

One possibility is that inclusion in the S&P 500 makes a stock more liquid and that this anticipated liquidity improvement can raise price.

Interpretation. The authors acknowledge that this mechanism could explain part of the initial rise, but they argue it cannot explain why the price increase later reverses while trading volume remains permanently elevated.

4.8 What the paper can and cannot distinguish

- The paper is very strong on distinguishing temporary pressure from permanent information effects.
- It is less able to quantify precisely how much of the temporary effect comes from inventory risk, transaction costs, or other short-run liquidity frictions.
- The deletion sample is too small to offer a parallel reversal analysis.

Interpretation. The paper identifies the presence of temporary price pressure much more cleanly than it identifies the exact microstructure channel behind that pressure.

4.9 Relation to the literature

Earlier studies of block trades, secondary distributions, and seasoned equity issues found immediate price effects but could not cleanly separate price pressure from adverse information. This paper's main contribution is to use an event where informational confounding is much weaker.

Interpretation. The paper is important because it improves *identification*, not because it introduces a more complicated econometric method.

5 Tables and figures

This paper is essentially **table-driven**. The published article has no central figure carrying the empirical argument. The evidence is concentrated in Tables I–VI plus the textual discussion of clean deletions.

5.1 Table I: post-announcement trading volume for additions

Read:

- In the full 1973–1983 additions sample, the mean day-1 volume ratio is 1.89 and the days-1-to-5 ratio is 1.29.
- In 1973–1977, the day-1 ratio is only 1.21 and statistically weak.
- In 1978–1983, the day-1 ratio jumps to 2.81 and the days-1-to-5 ratio is 1.66, both highly significant.

- In the late sample, 86% of firms have day-1 volume ratios above 1.

Economic message. The announcement clearly triggers abnormal trading, and the size of the effect rises dramatically once index funds become important. This is exactly what the demand-shift story predicts.

Table 1
Mean Post-Announcement Increases in Trading Volumes for Securities Added to the S&P 500 List

Period	N	Day 1 ^a				Days 1 to 5			
		MVR ^b	STD ^c	t ^d	Percent > 1	MVR	STD	t	Percent > 1
1973-83	194	1.89	2.63	4.65	55	1.29	0.88	4.56	54
1973-77	110	1.21	2.69	0.81	33	1.01	0.81	0.18	36
1978-83	84	2.81	2.27	7.16	86	1.66	0.84	7.14	77
1978-80	34	1.87	1.11	4.51	82	1.47	0.74	3.71	68
1981-83	50	3.45	2.63	6.46	90	1.79	0.88	6.23	83
1973	20	1.00	0.84	-0.00	35	0.96	0.51	-0.37	35
1974	12	1.22	1.90	0.41	25	0.83	0.85	-0.68	17
1975	18	1.15	6.31	0.77	17	1.45	1.61	1.19	44
1976	52	0.95	0.67	-0.59	38	0.90	0.39	-1.89	35
1977	8	1.28	1.32	0.60	38	1.19	0.53	0.98	63
1978	11	1.76	0.84	2.85	80	1.65	0.91	2.37	64
1979	14	1.92	1.43	2.42	79	1.31	0.70	1.64	64
1980	9	1.92	0.89	3.08	89	1.52	0.50	2.62	78
1981	19	4.72	2.99	5.42	89	2.12	1.09	4.50	89
1982	22	2.28	1.93	3.04	86	1.52	0.69	3.41	71
1983	9	3.51	2.04	3.48	100	1.73	0.53	3.94	100

^a Day 1 is the first day after announcement of the change.

^b Mean volume ratio. The cross-security mean of the ratio of volume in security *i* on day 1 to the average volume in that security in the 8 weeks preceding the week of the addition announcement, adjusted for splits and market volume. The expected value of this ratio will be equal to 1 if the announcement has no effect on volume.

^c The sample standard deviation of the volume ratios.

^d t-statistic for testing whether the mean of the volume ratios is different from 1.

Table II
Assets of Two Public Index Funds
(By Year)

Year	Vanguard (\$million)	Wells Fargo (\$million)
1974	— ^a	170
1975	— ^a	152
1976	14	439
1977	21	1,049
1978	66	1,718
1979	78	2,686
1980	98	4,200
1981	91	4,344
1982	110	5,100
1983	234	6,130

Note: The Vanguard Index Trust is a publicly traded mutual fund, the Wells Fargo S&P Fund is managed by the trust department of that bank.

^a The Vanguard Index Trust was founded in August 1976.

5.2 Table II: assets of two public index funds

Read:

- The Wells Fargo S&P Fund grows from \$170 million in 1974 to \$6.13 billion in 1983.
- The Vanguard Index Trust, founded in 1976, grows from \$14 million in 1976 to \$234 million in 1983.

Economic message. Table II is not a regression table. It supplies the key institutional fact behind the paper's period split: mechanical index-fund demand was far more important in the second half of the sample.

5.3 Table III: immediate excess returns for additions

Read:

- In the full 1973-1983 sample, the average day-1 excess return is 1.52% and the days-1-to-5 excess return is 1.62%.
- In 1973-1977, the day-1 excess return is only 0.21% and statistically insignificant.
- In 1978-1983, the day-1 excess return is 3.13% and the days-1-to-5 excess return is 2.77%.
- In the late sample, 96% of additions have a positive day-1 excess return.

Economic message. Prices move immediately after additions, but only in the era when index

funds are large. That pattern is difficult to reconcile with a stable information effect and strongly supports the demand-pressure interpretation.

Table III
Mean Excess Returns on the First Day Following an Announcement of an Addition to the S&P 500 List

Period	N ^a	Day 1				Days 1 to 5			
		Mean ^b	STD	t ^c	Percent > 0	Mean ^b	STD	t ^c	Percent > 0
1973-83	194	1.52	2.41	8.99	71	1.62	4.69	4.73	64
1973-77	110	0.21	1.89	0.99	53	0.69	4.85	1.45	53
1978-83	84	3.13	1.96	13.95	96	2.77	4.23	6.00	77
1978-80	34	2.97	1.79	8.84	97	2.74	3.40	4.69	85
1981-83	50	3.25	2.07	10.65	96	2.79	4.75	4.16	72
1973	20	0.82	2.93	1.58	65	1.08	7.35	0.66	50
1974	12	0.19	1.87	0.15	50	0.25	2.69	0.32	58
1975	18	-0.02	1.82	0.03	50	3.20	6.21	2.19	72
1976	52	-0.07	1.27	-0.28	40	-0.43	2.70	-1.09	46
1977	8	0.89	1.75	1.20	75	1.15	4.66	0.70	50
1978	11	2.91	1.72	5.18	91	3.15	4.04	2.59	91
1979	14	2.30	1.56	4.32	100	1.82	3.22	2.11	71
1980	9	4.07	1.85	6.70	100	3.66	2.79	3.94	100
1981	19	3.89	1.28	11.98	100	2.59	4.57	2.47	74
1982	22	2.59	2.40	4.97	91	2.99	5.66	2.48	68
1983	9	3.52	2.32	5.58	100	2.74	2.59	3.17	78

^a The number of securities in this table differs slightly from Table I because of data availability.
^b Mean excess return on day 1 in percent. The CRSP value-weighted index was used as the market proxy.

^c t -statistic for testing whether the mean of the excess returns on day 1 is different from zero. These t -statistics were computed after the excess returns were divided by the root-mean-squared error of the market model regression to adjust for heteroskedasticity.

Table IV
Mean Cumulative Excess Returns From Day 2 to Day T for the 84 Securities Added to the S&P 500 List in 1978-1983, and Two Tests of Whether the Negative of the Mean of the Cumulative is Equal to the Mean of the Day 1 Excess Return

Day 2 to Day T	Mean Cum. ^a	STD ^b	t -Test ^c		Posterior Odds Ratio ^d	Percent > 0	
			t	p -Value		Day 2 to Day T	Day 1 to Day T
2	0.09	1.76	11.10	0.0001	2.22×10^{32}	49	90
3	-0.11	2.99	7.63	0.0001	1.42×10^{17}	50	86
4	-0.09	3.45	6.88	0.0001	6.91×10^{13}	44	86
5	-0.36	3.80	5.75	0.0001	3.44×10^9	43	77
6	-0.69	3.89	4.91	0.0001	3.28×10^6	39	75
7	-0.64	4.34	4.55	0.0001	1.32×10^5	39	74
8	-0.66	4.49	4.32	0.0001	2.12×10^4	48	73
9	-0.97	4.58	3.64	0.0004	1.48×10^4	39	68
10	-1.18	4.75	3.17	0.0020	6.29×10^3	37	60
11	-1.74	5.29	1.95	0.0537	6.1576	37	56
12	-2.10	5.50	1.31	0.1933	1.7909	36	55
13	-2.26	5.70	1.01	0.3160	0.3207	36	58
14	-2.25	5.74	1.00	0.3213	0.5907	36	62
15	-2.25	5.96	0.95	0.3424	0.1910	32	57
16	-2.26	6.53	0.85	0.3947	0.0396	36	60
17	-2.60	6.53	0.38	0.7041	0.0092	32	56
18	-2.86	6.93	0.05	0.9575	0.0034	30	49
19	-2.83	6.88	0.09	0.9280	0.0020	31	50
20	-2.43	7.55	0.54	0.5910	0.0479	39	50
21	-2.49	7.43	0.48	0.6354	0.0222	32	54
22	-2.47	7.93	0.46	0.6444	0.0657	35	49
23	-2.61	7.82	0.32	0.7521	0.0552	37	50
24	-2.46	7.88	0.46	0.6484	0.0490	35	51
25	-2.52	7.36	0.41	0.6812	0.0539	35	55
26	-2.72	7.34	0.19	0.8475	0.0144	35	49
27	-2.63	7.79	0.29	0.7754	0.0120	32	46
28	-2.34	8.04	0.57	0.5702	0.0351	39	51
29	-2.39	8.20	0.50	0.6175	0.0415	37	50
30	-2.49	8.48	0.40	0.6935	0.0307	33	48

^a Mean cumulative excess return, in percent, from day 2 to the day indicated.

^b Sample standard deviation of the cumulative excess returns.

^c t -test for the equality of the means of two samples. The two samples are the day 1 excess returns of the added securities and the negative of the corresponding day 2 to day T cumulative excess returns. This test assumes that the two samples have different variances. Satterthwaite's [7] approximation for the degrees of freedom was used. In all cases, it was greater than ninety. The p -level presented is for a two-sided test. Note that the various t -statistics are not independent.

^d The posterior odds ratio for discriminating between

$H_0: \phi_i = 0$ for all i (no reversal)
 $H_1: \phi_i = -\mu_i$ for all i (full reversal)

in the following model:

$ER_{it} = \mu_i + \epsilon_i$
 $CER_{iT} = \phi_i + \eta_i$
with $\epsilon_i \sim N(0, \sigma^2)$
 $\eta_i \sim N(0, (T-1)\sigma^2)$, independently of ϵ_i

where ER_{it} is the excess return for the i^{th} security on day t , and CER_{iT} is the cumulative excess return in the i^{th} security from day 2 to day T . A low odds ratio favors the full-correction hypothesis. Note that the various odds ratios are not independent across T .

5.4 Table IV: cumulative reversals and full-correction tests

Read:

- In 1978-1983, the cumulative excess return from day 2 to day 11 is -1.74% and from day 2 to day 21 is -2.49%.
- By day 11, 63% of firms have negative cumulative post-day-1 returns; by day 21 the fraction is 68%.
- From day 10 onward, the two-sample t -test cannot reject full reversal at the 5% level.
- For all $T > 15$, the posterior odds of no reversal relative to full reversal are below 0.07.

Economic message. This is the most decisive table in the paper. The initial price rise is not permanent. The data favor temporary pressure with correction, not a lasting new equilibrium price.

5.5 Table V and Table VI: anticipation, immediate dynamics, and longer-run volume

Table V
Mean Increases in Daily and Weekly Trading Volumes Around the Announcement Date of the 84 Additions to the S&P 500 List in 1978–1983

Period	MVR ^a	STD	<i>t</i> ^b	Percent > 1
Days				
Day -2	0.86	0.80	-1.51	31
Day -1	0.89	0.74	-1.36	35
Day 0 (Wed.)	0.95	0.94	-0.53	26
Day 1	2.81	2.27	7.16	86
Day 2	1.63	1.66	3.44	54
Day 3	1.50	1.35	3.33	57
Day 4	1.21	0.88	2.13	46
Day 5	1.16	0.81	1.73	48
Day 6	1.23	1.23	1.67	38
Day 7	1.20	1.44	1.25	37
Days 1 to 5 (Thurs. to Wed.)	1.66	0.84	7.14	77
Monday to Friday Weeks				
Week 8	0.98	0.44	-0.44	38
Week 7	0.96	0.43	-0.78	40
Week 6	1.02	0.46	0.46	51
Week 5	1.04	0.49	0.68	44
Week 4	1.06	0.52	1.14	49
Week 3	1.09	0.57	1.52	52
Week 2	0.94	0.37	-1.42	40
Week 1	0.90	0.40	-2.33	33
Week 0 ^c	1.46	0.76	5.57	75
Week 1	1.25	0.65	3.57	60
Week 2	1.15	0.82	1.72	46
Week 3	1.14	0.67	1.95	49
Week 4	1.14	1.06	1.18	44
Week 5	0.98	0.53	-0.40	39
Week 6	1.06	0.67	0.86	43
Week 7	1.18	0.98	1.70	50
Week 8	1.14	0.70	1.85	51

^a Mean volume ratio. The cross-security mean of the ratio of volume in security *i* to the average volume in that security in the 8 weeks preceding the week of the addition announcement, adjusted for splits and market volume. The expected value of this ratio will be equal to 1 if the announcement has no effect on volume.

^b *t*-statistic for testing whether the mean of the volume ratios is different from 1.

^c The weekly data are measured from Monday to Friday. Since announcements of changes in the S&P 500 are always made after trading closes on Wednesday, week 0 is composed of three pre-announcement days (days -2 to 0) and 2 post-announcement days (days 1 and 2). Week 1 is composed of days 3 to 7.

Table VI
Mean Excess Returns Around the Announcement Date for the 84 Securities Added to the S&P 500 List in 1978–1983

Event Day	Mean ^a	STD	<i>t</i> ^b	Percent > 0
-10	-0.73	2.23	-3.10	35
-9	-0.04	2.12	-0.44	44
-8	-0.15	1.54	-0.99	44
-7	-0.17	1.58	-1.12	37
-6	-0.06	1.90	-0.25	44
-5	-0.31	2.19	-1.05	49
-4	0.21	2.05	1.15	50
-3	0.11	1.50	0.42	44
-2	0.01	1.76	0.12	46
-1	-0.13	1.42	-0.39	45
0	0.01	2.15	0.53	46
1	3.13	1.96	13.95	96
2	0.09	1.77	0.84	49
3	-0.21	2.21	-0.39	45
4	0.02	1.95	0.14	46
5	-0.27	1.83	-1.10	42
6	-0.33	1.94	-1.58	40
7	0.04	1.74	0.24	56
8	-0.01	1.81	0.23	43
9	-0.31	1.72	-1.49	38
10	-0.19	2.00	-1.03	45
11	-0.59	1.62	-3.29	36
12	-0.36	1.81	-2.09	45
13	-0.17	1.81	-0.93	44
14	0.01	1.87	0.21	46
15	-0.00	1.93	0.22	46
16	-0.03	2.31	-0.07	50
17	-0.35	1.50	-2.05	38
18	-0.27	1.80	-0.86	39
19	0.05	1.91	0.12	46
20	0.39	2.44	0.77	48
21	-0.03	1.89	0.23	43
22	0.00	2.10	0.14	39
23	-0.12	2.13	-0.62	48
24	0.16	1.81	0.89	46
25	-0.01	1.85	-0.16	45
26	-0.19	1.75	-0.39	44
27	0.09	2.26	-0.05	49
28	0.29	1.84	1.83	61
29	-0.05	1.97	-0.24	48
30	-0.10	21.7	-0.04	42

^a Mean excess return in percent. The CRSP value-weighted index was used as the market proxy. Cumulatives are presented in Table IV.

^b *t*-statistic for testing whether the mean of the excess returns on day *t* was different from zero. These *t*-statistics were computed after the excess returns were divided by the root-mean-squared error of the market model regression to adjust for heteroskedasticity.

Read:

- Table V shows no meaningful pre-announcement volume buildup in 1978–1983: days -2 to 0 have mean volume ratios of 0.86, 0.89, and 0.95.
- Day 1 volume is 2.81, then falls quickly toward normal levels; by day 7 the mean ratio is 1.20.
- After week 0, weekly volume ratios mostly lie between about 0.98 and 1.18.
- The paper also reports that trading volume remains permanently higher in a later quarter, with a mean ratio of 1.26.
- Table VI shows that pre-announcement excess returns are mostly insignificant, while day 1 excess return is 3.13% and day 11 return is -0.59%.

Economic message. There is little evidence of anticipation, a very large immediate reaction, a subsequent price giveback, and a persistent increase in trading volume. The last fact is important

because it suggests that inclusion changes the stock's investor base permanently even though the price impact is temporary.

5.6 Opening-price evidence inside the late sample

Read:

- For 18 firms in a smaller 1981–1983 transactions subsample, the mean return from day-0 close to day-1 open is 1.45%.
- The mean open-to-close day-1 return is another 1.01%.

Economic message. Much of the adjustment happens immediately after the public announcement and before the next opening trade, which fits the idea that the market quickly capitalizes the demand imbalance.

5.7 Deletions evidence (Section II.D)

Read:

- In the small clean deletions sample, the mean day-1 volume ratio is 4.92 and the days-1-to-5 ratio is 3.59.
- The day-1 excess return is -1.4% and statistically significant.
- Because the sample has only 13 events and several are clustered in utilities, the paper does not make strong claims about subsequent reversal.

Economic message. The sign of the deletion response lines up with price pressure, adding breadth to the additions evidence even if it is too small a sample for a definitive dynamic analysis.

6 Short summary

- The paper studies S&P 500 list changes as demand shocks that are plausibly free of new information about firm fundamentals.
- After additions are announced, trading volume jumps and prices rise immediately, especially in 1978–1983 when index funds are large.
- The day-1 price effect is then reversed over the next two to three weeks.
- This reversal is the central fact that favors the price-pressure hypothesis over both the efficient-markets view of no price effect and the imperfect-substitutes view of a permanent price change.
- Pre-announcement returns and volume show little sign of anticipation.
- The no-information interpretation is strengthened by S&P's stated criteria, weak demand for advance announcement information, and the fact that the effect appears mainly when index funds become economically important.
- Clean deletions also show the expected negative price response, though the sample is small.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that large demand shifts can move stock prices even when they appear not to convey new information, and that the resulting price impact is mostly temporary rather than permanent. In the context of S&P 500 additions, the evidence supports the price-pressure hypothesis: prices rise when index-fund demand arrives, then move back toward prior levels over the next few weeks.

7.2 Economic intuition

When many investors are forced to buy the same stock quickly, somebody else must temporarily absorb the order imbalance. Those passive suppliers of liquidity do so only at a better price. Once the immediate pressure passes, the extra compensation embedded in price is no longer needed, so the stock drifts back toward its full-information value. That is why the price effect is temporary while the change in who holds the stock can be more persistent.

7.3 Takeaway

The paper's message is not just that *demand matters*. It is that short-run demand can matter a lot even when long-run fundamentals do not change. By exploiting S&P 500 additions as unusually clean demand shocks, Harris and Gurel provide classic evidence that stock prices can display temporary price pressure when liquidity has to be supplied on short notice.